

You can use tridents enchanted with Riptide while riding entities

MC-222949 Player

Status	Resolved / Fixed
Affected	1.16.5, 21w14a, 21w15a, 21w16a, 21w17a, 21w18a (+30 more)
Fixed in	26.1 Snapshot 10
Reported	2021-04-11

STEPS TO REPRODUCE

1. Launch Minecraft and enter a world where commands can be used.
2. Set the weather to rain with `/weather rain``.
3. Give yourself a Riptide III trident with `/give @s minecraft:trident[minecraft:enchantments={"minecraft:riptide":3}]``.
4. Mount any entity.
5. Attempt to throw the Riptide-enchanted trident while riding the entity.
6. Observe that the trident can be used while riding the entity and may damage the entity being ridden.

OBSERVED BEHAVIOUR

While the player is riding another entity, a Riptide-enchanted trident can still be activated when the player is in rain or water. The activation propels the player as normal instead of being rejected because the player is a passenger. Depending on the mounted entity and the resulting movement, the Riptide activation can also damage the entity the player is riding.

EXPECTED BEHAVIOUR

While the player is riding an entity, attempting to use a Riptide-enchanted trident should fail, even during rain or while in water. The trident should not be thrown or propel the player, and the ridden entity should not be damaged as a result. Riptide tridents should remain usable normally when the player is not riding another entity.

ENVIRONMENT

Minecraft Java Edition Versions: 1.16.5, 21w14a, 21w15a, 21w16a, 21w17a, 21w18a, 21w19a, 21w20a, 1.17 Pre-release 4, 1.17 Pre-release 5, 1.17 Release Candidate 1, 1.17, 1.17.1 Pre-release 1, 1.17.1 Pre-release 2, 1.17.1 Pre-release 3, 1.17.1 Release Candidate 1, 1.17.1, 21w37a, 1.18.1, 22w05a, 22w06a, 1.18.2, 1.19, 1.19.1, 1.19.2, 22w43a, 1.19.3, 1.19.4, 1.20.1, 24w20a, 1.21, 1.21.3, 1.21.4, 1.21.7, 25w41a, 1.21.11 (includes snapshots)

ORIGINAL DESCRIPTION

The Bug:

You can use tridents enchanted with riptide while riding entities and in some cases, this can damage the entity you're riding.

Steps to Reproduce:

- Set the weather to "rain" and give yourself a trident enchanted with riptide by using the command provided below.
`/give @s minecraft:trident[minecraft:enchantments={"minecraft:riptide":3}]`
- Ride any entity and attempt to throw the riptide trident.
- Take note as to whether or not you can use tridents enchanted with riptide while riding entities.

Observed Behavior:

You can use riptide tridents.

Expected Behavior:

You would not be able to use riptide tridents.

Code Analysis:

Code analysis by accountid:unknown can be found below.

The following is based on a decompiled version of Minecraft 1.19.2 using MCP-Reborn.

```

net.minecraft.world.item.TridentItem.java
public class TridentItem extends Item implements Vanishable {
...
public InteractionResultHolder<ItemStack> use(Level level, Player player, InteractionHand interactionHand) {
ItemStack itemstack = player.getItemInHand(interactionHand);
if (itemstack.getDamageValue() >= itemstack.getMaxDamage() - 1) {
return InteractionResultHolder.fail(itemstack);
} else if (EnchantmentHelper.getRiptide(itemstack) > 0 && !player.isInWaterOrRain()) {
return InteractionResultHolder.fail(itemstack);
} else {
player.startUsingItem(interactionHand);
return InteractionResultHolder.consume(itemstack);
}
}
...

```

If we look at the above class, we can see that there are two checks that are carried out before allowing the player to use a trident enchanted with riptide. One of these is to check if the player is actually holding a trident enchanted with riptide and the other is to check if they are in water or rain. The game doesn't check if the player is currently riding an entity before allowing them to throw a trident enchanted with riptide in the rain, therefore resulting in this problem occurring.

Fix:

Simply altering the appropriate existing "if" statement within this piece of code to check if the player is a passenger before allowing them to throw a trident enchanted with riptide in the rain, will resolve this problem. We can achieve

MANUAL RESULT

Version used	
Reproduced?	YES / NO / PARTIAL
Crash file	
Notes	
Tested by / date	

Live issue: <https://bugs.mojang.com/browse/MC/issues/MC-222949>